

Co5 AT3

1. Suppose you are working on an industrial monitoring system. How would you use Bayesian Network to predict machine failures?

A Bayesian Network is Directed Acyclic graph which represent nodes as variables and paths as probabilistic dependencies.

For industrial monitoring system, i would represent factors such as temperature, pressure, vibration, motor current and machine failure as nodes.

Ex: Temperature $\rightarrow$ overheating $\rightarrow$ machine failure

The network would be trained using historical sensor data. When new sensor readings arrive, such as unusually high temperature and vibrations as i would enter these observations into Bayesian network.

3. If company wants to build speech recognition system, how would you apply HMM?

A HMM is suitable for speech recognition cause speech is sequential signal and actual linguistic states are directly observable.

The audio signal would be first divided into short frames. From each frame, features such as MFCCs can be extracted.

Ex: Audio $\rightarrow$ MFCC Feature $\rightarrow$ HMM $\rightarrow$ Phoneme Sequence $\rightarrow$ words

When new audio sequence is given, algorithm such as viterbi algorithm, such can determine the most likely sequence of hidden speech state.

6. In healthcare diagnosis problem, how would you compose directed and undirected for representing symptom relationships?

1. Directed Graphical model - Bayesian network

* They use directed edges.

Ex:

Smoking $\rightarrow$ Lung disease $\rightarrow$ Cough

They are useful when we want to represent casual or directional relationship.

They use conditional probability distributions and conditional independence based on graph structure.

2. Undirected model - MRF

* They use undirected edges.

They are useful when relationships are symmetric.

8. How would you use HMM to analyze customer purchase sequence in e-commerce recommendation system?

customer purchase are naturally sequential, so an HMM can model their behavior over time.

Ex:

Browsing $\rightarrow$ interested $\rightarrow$ purchase $\rightarrow$ intent $\rightarrow$ purchase

The model can infer that hidden state is like high purchase intent.

The recommendation system can then recommend products related to laptops, accessories, warranties or complementary products.